Category	Structure / Algorithm	Simple Real-World Example
Searching	Binary Search (Array + Sort)	Finding a word in a dictionary by flipping to the middle, then cutting the search in half.
	Binary Search (Linked List)	Trying to find the middle car of a train by walking through every single car first.
	Linear Search (Array)	Checking a short row of mailboxes one by one from left to right.
	Linear Search (Linked List)	Following a scavenger hunt where each clue gives you the location of the next clue.
Linked Lists	Singly Linked List	A basic music playlist where you can only click "Next", never "Previous".
	Doubly Linked List	A web browser's "Back" and "Forward" buttons for navigating your page history.
	Circular Linked List	A Monopoly game board where passing "Go" loops you back to the beginning.
Stacks	Stack (Array)	A spring-loaded stack of trays in a cafeteria. The last tray put on top is the first one taken.
	Stack (Linked List)	The "Undo" button in a text editor that keeps remembering your actions without a hard limit.
Queues	Linear Queue (Array)	A fast-food drive-thru lane that only has physical space for 5 cars.
	Linear Queue (Linked List)	A customer service phone queue that can

		keep adding callers on hold endlessly.
	Circular Queue	A car dashcam that continuously records, automatically overwriting the oldest footage when full.
	Double Ended Queue (Deque)	A row of seats in a stadium where people can enter or leave from both the left and right aisles.
Trees & Logic	Tree	Your computer's file system (Folders containing sub-folders containing files) or a Family Tree.
	Factorial Recursion	Opening a set of Russian nesting dolls, one by one, until you reach the solid doll in the center.
	Tower of Hanoi	Stacking different-sized rings on pegs, with the strict rule that a larger ring can never sit on a smaller one.